

Identifying When Steady-State Flow Simulations In Patient-Specific Coronaries Recapitulate Pulsatile Flow Dynamics

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Abstract—Computational models have emerged as a powerful tool to non-invasively monitor vital biomarkers in coronary artery disease, providing a viable alternative to conventional invasive methods and improving clinical decision-making. The precision of these *in silico* models, however, is often counterbalanced by their substantial computational cost. Pulsatile flow conditions, which closely mimic physiological conditions, are typically employed in computational fluid dynamics (CFD) simulations to capture dynamic changes in hemodynamic variables, but come at a significant computational cost. This study addresses the critical question of the necessity for complex pulsatile models versus the adequacy of simpler steady-state models to capture hemodynamic metrics such as fractional flow reserve, velocity, vorticity, and wall shear stress within the cardiac cycle. By comparing steady-state and pulsatile flow simulations in 12 patients with stenosed coronary arteries, our research evaluates whether steady-state simulations can accurately reflect patient-specific hemodynamic profiles at key clinical moments. The results indicate a comparability of metric magnitudes and distributions between the two types of simulation, particularly during diastole, with minimal influence from the choice of inlet waveform. This suggests that for metrics such as fractional flow reserve (FFR), steady-state simulations are sufficiently accurate and markedly reduce computational load, whereas pulsatile simulations may be necessary for capturing complex dynamics at systole. This distinction underscores the potential for selective application of steady-state models in clinical practice, facilitating the integration of computational fluid dynamics modeling by identifying when the reduced complexity of steady-state simulations can effectively support patient care in coronary artery disease.

I. INTRODUCTION

Substituting steady-state (SS) inlet flow simulations in personalized computational fluid dynamics (CFD) models in lieu of pulsatile state (PS) models can improve the applicability of CFD modeling in clinical settings by reducing computational complexity. In the last decade, CFD modeling has grown rapidly and has found clinical applicability to help diagnose coronary artery disease (CAD) [1]. CFD models have the capacity to calculate virtual biomarkers that are correlated to CAD, although most simulations that compute these metrics rely on pulsatile flow at the inlet [2]. PS inlet boundary flow

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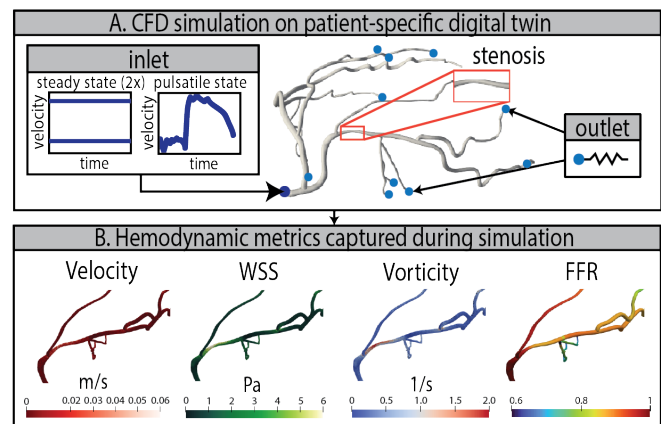


Fig. 1. A. The rigid mesh of a patient's coronary tree determines wall conditions for a CFD simulation. A resistance model is simulated at all outlets, capturing resistance from distal branches. Steady-state and pulsatile-state inflows were given by constant flow rates and a patient-scaled waveform, respectively. Within the coronary tree is a stenosis annotated by an expert cardiologist. B. At a fixed time point, velocity, WSS, and vorticity are captured at each fluid point across the geometry, but only points along the stenosis were extracted for post-processing. FFR was calculated across the cardiac cycle at an annotated point more distal than the end of the stenosis. Proximal and distal ends of the stenosis were annotated manually, along with the minimum lumen diameter.

encapsulates fluctuations in blood flow throughout a cardiac cycle to preserve physiological accuracy. However, these methods result in extended model convergence times compared to SS models. In order to circumvent these issues while maintaining the physiological accuracy and spatial nature of 3D models, simulations can instead be initialized by a constant inflow of blood, derived from specific time points within the pulsatile waveform. With faster simulations [3], a fixed flow rate can increase clinical translatability, but only if it can provide hemodynamic information equivalent to PS data.

Addressing the question of where PS is necessary as opposed to SS represents a significant area of interest within CAD research, filled with conflicting conclusions. Results vary from differences in idealized geometries, sampling points,

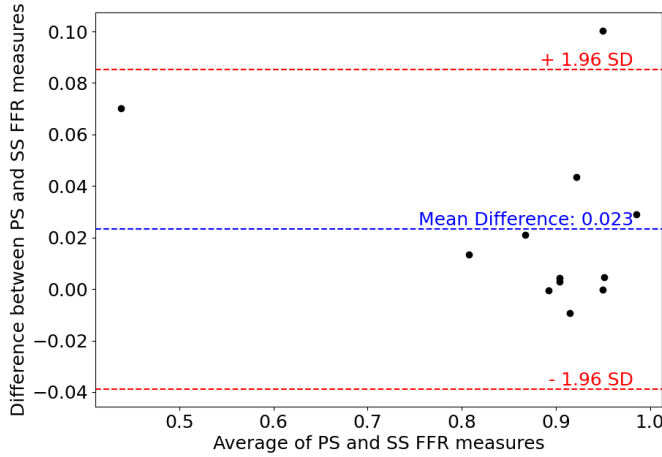


Fig. 2. FFR was computed across the entire cardiac cycle in a patient specific manner for PS simulations, and at simulated end-diastole conditions for SS patients ($n = 12$). Quantitative comparison between FFR calculated among the same patients using PS and SS inlet flow conditions are shown.

hemodynamic parameters, and model design [4]–[8]. For example, the choice of SS vs. PS inlet conditions has been shown to have a minimal effect on the distribution of wall shear stress in intracranial aneurysms [3], but for stenosed coronary arteries, SS simulations underestimate pressure, velocity, and wall shear stress (WSS) [9]. Comprehensive comparisons that span conventional flow metrics such as pressure and velocity to more complex metrics like vorticity in 3D patient-specific anatomies are needed. This study aims to evaluate multiple parameters within a single patient cohort, determining the effectiveness of SS simulations in accurately reflecting a wide range of biomarkers critical to clinical practice.

The current literature suggests that atherosclerotic lesions develop in areas with low WSS, while regions with persistent high WSS are atheroprotective [10], [11]. Beyond WSS, there are other hemodynamic metrics that could be used to evaluate and monitor CAD progression. The clinical standard to gauge the severity of CAD is the fractional flow reserve (FFR), a pressure wire-based index calculated by invasive coronary angiography to assess the severity of coronary stenoses and, subsequently, the need for intervention [12]. CFD-derived FFR has the benefits of assessing stenosis severity without the drawbacks that limit the use of the invasive method, and simulations with SS and PS inlet flow conditions have been validated against clinical measurements [1], [13]. Although there are CFD-derived FFR algorithms that have FDA approval [14], clear comparisons between PS and SS approaches within the same cohort of patients remain elusive. Additionally, FFR does not capture local flow patterns, indicating the potential value of metrics such as vorticity, which quantifies rotational tendency in a flow field. Vorticity has been correlated to FFR and early indications point to it providing further insight into the ischemic burden of a lesion [15]–[18]. However, there is a noticeable gap in research comparing PS and SS simulations

for vorticity calculations, largely due to its nascent application in clinical settings. Our study aims to bridge this gap by exploring the effect of pulsatility on an array of hemodynamic metrics.

We propose that in cases where dynamic flow rate and inertia effects are unnecessary, using steady-state boundary conditions as an alternative can render CFD modeling more applicable in clinical fields. The effectiveness of SS simulations in mirroring the results of PS simulations varies depending on factors such as the geometry of the computational domain, the timing within the cardiac cycle, and the specific hemodynamic metric being examined. Our work investigates how inlet waveforms affect the simulation of blood flow in 3D patient-specific models of stenosed coronary arteries, focusing on FFR, velocity, WSS, and vorticity. The work not only builds on previous findings by applying them to more realistic 3D anatomies and boundary conditions, but it will also be the first to explicitly compare PS vs. SS simulations on vorticity and FFR in a single patient cohort. As their agreement could vary across a geometry and since clinicians require data in stenotic regions, our analysis will cover clinically annotated stenoses. Our results indicate that at specific time points in the cardiac cycle, SS simulations can effectively replace PS simulations for an accurate assessment of key hemodynamic indicators.

II. METHODS

A. Clinical data collection

Our dataset consisted of anonymized, blinded data from 12 patients with CAD lesions, measured by invasive angiography [19]. The protocol was approved by the Massachusetts General Brigham Institutional Review Board (IRB Protocol #2015P001084). More details on patient angiograms and reconstruction methods can be found in [20]. Angiogram images were captured at end-diastole, with the location of the stenosis identified by an experienced cardiologist.

B. Fluid algorithm and boundary conditions

Each 3D coronary geometry was reconstructed from a pair of angiograms using the algorithm described in [21]. Outlet boundary conditions were defined with patient-tuned total coronary microresistance. Inlet waveforms have constant or pulsatile flow rates that define the SS and PS simulations, as shown in Fig. 1. SS inlet flow rates were taken from a pulsatile input waveform at systole and peak-diastole. These measurements were patient-tuned using heart rate and cardiac output (CO). Blood was modeled as an incompressible Newtonian fluid with $\rho = 1,060 \text{ kg/m}^3$. CFD simulations were performed with HARVEY, a massively parallel hemodynamics solver based on the lattice Boltzmann method (LBM), which simulated cardiac cycles of blood flow along the entire coronary artery tree for each patient [22]. We computed a patient-specific dynamic viscosity that was tuned according to clinically measured hematocrit as defined in [23]. We evaluated temporal and spatial convergence and found that one

cardiac cycle was sufficient, so the second cycle was used for analysis.

C. Evaluation of hemodynamic metrics

WSS was calculated as in [24]. Vorticity was calculated at each point in the geometry as the magnitude of curl in the velocity vector field. Given the velocity vector $\vec{v}$ at each point:

$$||\nabla \times \vec{v}|| = \sqrt{\left(\frac{\partial v_z}{\partial y} - \frac{\partial v_y}{\partial z}\right)^2 + \left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right)^2 + \left(\frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}\right)^2}. \quad (1)$$

FFR is defined as

$$FFR = \frac{p_{distal}}{p_{proximal}} \quad (2)$$

where p is physical pressure. To calculate physical pressure from LBM simulation results (e.g. density), values at both systole and diastole are required [25]. Data was extracted at those time points for PS simulations, while two SS simulations were run with systole and diastole flow rates to best resemble PS simulations at the chosen time points. We took the average density at the two time points to calculate the SS FFR.

D. Eliminating Erroneous Slices

Simulations were performed for the entire 3D domain of each coronary tree, but we restricted our analysis to the region of interest, *i.e.* the stenosis. Stenosis lengths vary across a population; thus, end points of a patient's stenosis were manually identified as locations where the vessel diameter is still, or returns to, normal. The primary slice is at the minimal lumen diameter (MLD) of the stenosis, around which the slices were taken in both the proximal and distal directions until the ends were reached. We restricted the analysis to main branches and excluded data from side branches, particularly when a stenosis is found near a bifurcation.

III. RESULTS

A. FFR ratios across patient stenotic region

Computed FFR values were found to be highly similar in both PS and SS simulations across our patient cohort. The mean difference between the FFR from the PS and SS simulations, as shown in Fig. 2, was 0.023 with a standard deviation of 0.0316. The percent discrepancy in the calculated FFR values averaged in all cases was 3.06% between the PS and SS simulations with a standard error of 1.28.

B. Pulsatile-state vs. steady-state spatial distributions

Curves tracking the evolution of hemodynamic biomarkers were graphed throughout the stenosis region for a representative subset of individual patients, the results of which are shown in Fig. 3. PS and SS simulation biomarkers are displayed on the same graph at both the diastole and systole time points, showing qualitatively similar shapes along the duration of the stenosis region. When treating PS simulations as ground truth, the average percent difference ($\pm SE$) between the magnitudes of the SS and PS waveforms at the diastole time point was determined to be $4.43\% \pm 0.59$ for vorticity, $3.67\% \pm 0.63$ for velocity, and $6.07\% \pm 0.99$ for WSS. For the percent difference between the SS and PS waveforms at

the systole time point, we found $19.27\% \pm 1.01$ for vorticity, $18.94\% \pm 1.0$ for velocity, and $23.34\% \pm 1.46$ for WSS. To compare the degree of similarity in shape of the metric waveforms, we scaled all waveforms to have an integral of 1 such that waveform shape similarity can be analyzed absent of variance in magnitude. We found that comparing SS systole to PS systole waveforms within patients and then averaging across the cohort led to a percent difference ($\pm SE$) of $2.29\% \pm 1.44$ for vorticity, and comparing SS diastole with PS diastole led to $1.29\% \pm 0.31$. For velocity, the comparison of the systole time points resulted in $2.24\% \pm 1.8$ and $0.5\% \pm 0.13$ for diastole. Finally, for WSS, the shape comparison resulted in $4.65\% \pm 2.43$ for systole and $2.59\% \pm 0.55$ for diastole.

IV. DISCUSSION AND CONCLUSION

CFD modeling provides a non-invasive, accurate approach for assessing hemodynamic markers crucial for understanding CAD progression. Despite their detailed insights, PS simulations are computationally intensive. Conversely, SS simulations offer a cost-effective alternative, producing hemodynamic insight with significantly lower computational demands, thus improving their clinical applicability.

Our study confirms the accuracy of SS simulations, particularly in diastolic phases, where they closely match PS simulations in calculating FFR values and other hemodynamic metrics with minimal deviation. This evidence supports the use of SS simulations as a cost-effective alternative for clinical assessments, effectively capturing the physiological conditions of various hemodynamic indicators during diastole.

Systolic flow analysis showed notable discrepancies in metrics for certain coronary geometries, suggesting that SS simulations may not fully capture the complexity of systolic hemodynamics as accurately as PS simulations, particularly in cases with complex anatomical features or severe stenosis. Although diastolic measures were less affected by patient-specific differences, indicating the robustness of SS simulations to assess high flow states, pronounced variations during systole highlight the importance of considering anatomical diversity and flow rates for accurate simulation results.

This insight identifies the conditions under which SS simulations can efficiently replace PS simulations, especially for diastolic analyses. Future research is needed to enhance SS simulations for systolic conditions, perhaps by developing more precise approximation methods or scaling factors for a wider range of cardiac phases. However, challenges such as the identification of visual stenosis, the limited analysis of time points, and the variability of coronary geometries due to disease must be overcome to improve the precision and clinical applicability of these simulations.

In conclusion, our study highlights the potential of SS simulations to accurately reflect key hemodynamic indicators such as WSS, velocity, FFR, and vorticity, serving as a streamlined alternative to the more computationally demanding PS simulations. Yet, the varied accuracy across diastolic and systolic phases, particularly with complex metrics and diverse patient geometries, reveals the critical influence of anatomical and

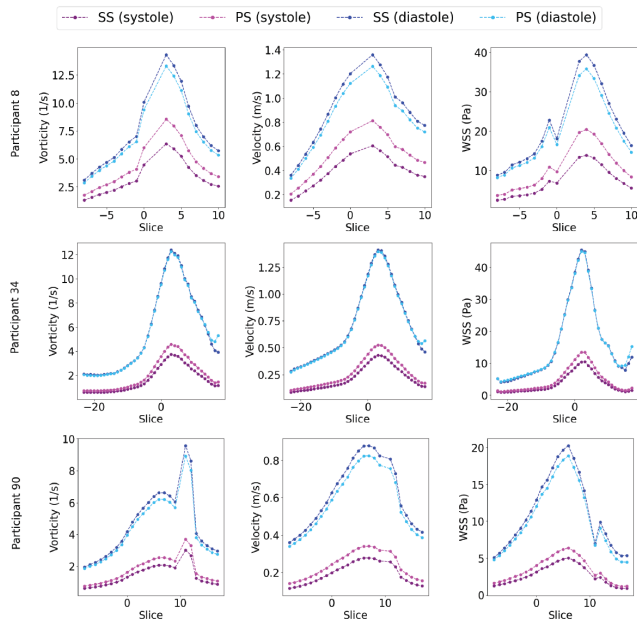


Fig. 3. Hemodynamic metrics of vorticity, velocity and WSS were tracked at each slice along the entire stenotic region for three representative patients. Patients were selected from the cohort based on variance between hemodynamic metrics, where participant 8 is the highest variance, participant 34 is of average variance and participant 90 is the lowest variance. Along the x -axis, 0 denotes the MLD slice, negative slice number represent number of slices proximal to the MLD, and positive values represent distal slices.

flow dynamics. This observation confirms that SS simulations cannot be universally substituted for PS simulations, indicating that under certain conditions – especially those involving systolic flows or patient-specific anatomical variations – PS simulations remain necessary to accurately capture the full spectrum of cardiovascular dynamics.

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